

17/11/2017(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
May - June 2017

**Max. Marks:100**

Class: S.Y. B.Tech

Name of the Course: Database Management Systems

Course Code: UITC302

**Duration: 3 Hours**

Semester: III

Branch: IT

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                   | Max. Marks |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q1 (a)       | Discuss database system architectures in detail.                                                                                                                                                                                  | 10         |
| Q1 (b)       | What are five main functions of a database administrator?                                                                                                                                                                         | 10         |
| Q2 (a)       | Explain any five domain constraints of DBMS with suitable example.<br><br><b>OR</b><br>Discuss the entity integrity and referential integrity constraints. Why is each Considered important?                                      | 10         |
| Q2 (b)       | Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.<br>Also map above E-R diagram to relational model.      | 10         |
| Q3 (a)       | What are the differences among primary, secondary, and clustering indexes? Which of the indexes are dense, and which are not?<br><br><b>OR</b><br>What is Static and dynamic hashing? How does it work? What are its limitations? | 10         |
| Q3 (b)       | Discuss the types of privileges at the relation level with the help of suitable SQL statements.<br><br><b>OR</b><br>What is the difference between discretionary and mandatory access control with reference to database?         | 10         |
| Q4 (a)       | What is database normalization? Discuss 1NF, 2NF, 3NF with suitable example.                                                                                                                                                      | 10         |

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|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q 4 (b) | <p>Consider the relation REFRIG(Model#, Year, Price, Manuf_plant, Color), which is abbreviated as REFRIG(M, Y, P, MP, C), and the following set F of functional dependencies:<br/> <math>F = \{M \rightarrow MP, \{M, Y\} \rightarrow P, MP \rightarrow C\}</math></p> <p>a. Evaluate each of the following as a candidate key for REFRIG, giving reasons why it can or cannot be a key:<br/> <math>\{M\}</math>, <math>\{M, Y\}</math>, <math>\{M, C\}</math>.</p> <p>b. Based on the above key determination, state whether the relation REFRIG is in 3NF and in BCNF, giving proper reasons.</p> | 10 |
| Q5 (a)  | <p>Consider the three transactions T1, T2, and T3, and the schedules S1 and S2 given below. Draw the serializability (precedence) graphs for S1 and S2, and state whether each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s).</p> <p>T1: r1 (X); r1 (Z); w1 (X);<br/> T2: r2 (Z); r2 (Y); w2 (Z); w2 (Y);<br/> T3: r3 (X); r3 (Y); w3 (Y);</p> <p>S1: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); w3 (Y); r2 (Y); w2 (Z); w2 (Y);<br/> S2: r1 (X); r2 (Z); r3 (X); r1 (Z); r2 (Y); r3 (Y); w1 (X); w2 (Z); w3 (Y); w2 (Y);</p>   | 10 |
| Q5(b)   | <p>Describe the shadow paging recovery technique. Under what circumstances does it not require a log?</p> <p style="text-align: center;"><b>OR</b></p> <p>How to test Conflict Serializability of a Schedule? What is View Equivalence and View Serializability</p>                                                                                                                                                                                                                                                                                                                                 | 10 |

20-11-2017 (E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
Nov - Dec 2017

Max. Marks: 100

Class: SY BTech

Name of the Course: Data Structures

Course Code: UUTC303

Duration: 3hrs

Semester: III

Branch: IT

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary
- (4) Use PEN for writing final answers

| Question No. |                                                                                                                                                                                                                                                                                            | Max. Marks |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q1 (a)       | Explain following operations along with ADT signature with respect to DLL with the help of suitable block diagrams.<br>i) InsertBefore                      ii) InsertAfter<br>iii) DeleteElement                  iv) DeleteAfter                                                         | 10         |
| Q1 (b)       | Write a C program to implement circular SLL which maintains TAIL pointer (pointer pointing to last node) only.<br><br>OR<br>Construct max-heap by performing following operations in the given order<br>Insert - 5, 9, 10, 0, 2, 6, 10, 24, 3, 34, 9, 8, 12, 14<br>Delete - 10, 6, 12      | 10         |
| Q2 (a)       | Write C program to implement linear queue using array.                                                                                                                                                                                                                                     | 10         |
| Q2 (b)       | Write a C program to implement static stack and use it in the implementation of palindrome checker function.                                                                                                                                                                               | 10         |
| Q3 (a)       | Identify suitable data structure to add two polynomials and write a C function to implement addition of polynomials using it.                                                                                                                                                              | 10         |
| Q3 (b)       | Write algorithm or pseudo code for prefix to postfix expression conversion. Apply it to convert the following<br><br>Prefix expression : * / 5 - 9 2 - + 14 23 / 24 3<br><br>OR<br>Show the use of appropriate data structure in the C program of prefix to postfix expression conversion. | 10         |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q3(c) | <p>Identify the data structure used in expression tree construction. Show its use by constructing expression tree for the following expression.</p> <p style="text-align: center;">Infix expression : <math>(5/(9-2)*((14+23)-(24/3)))</math></p>                                                                                                                                                                                                                                           | 10 |
| Q4(a) | <p>Construct the 3-way B-Tree by performing following operations on an empty tree in the given order.</p> <p style="text-align: center;">Insert - 5, 9, 10, 0, 2, 6, 10, 24, 3, 34, 9, 8, 12, 14<br/>Delete - 10, 6, 12</p> <p style="text-align: center;">OR</p> <p>Construct BST and AVL tree by inserting following numbers on an empty tree in the given order. Comment on the trees constructed.</p> <p style="text-align: center;">5, 9, 10, 0, 2, 6, 10, 24, 3, 34, 9, 8, 12, 14</p> | 10 |
| Q4(b) | <p>Explain the different ways to represent the graphs in computer. Depict the use of queue in the graph traversal with the help of an example.</p>                                                                                                                                                                                                                                                                                                                                          | 10 |
| Q5    | <p>i) What is the most efficient search technique among those you have studied? Why it is most efficient?</p> <p>ii) Differentiate between selection sort and insertion sort</p> <p style="text-align: center;">OR</p> <p>Hash the following keys into a hash table of size 14 using standard hash function. Handle the collisions using quadratic hashing.</p> <p style="text-align: center;">Key - 5, 9, 10, 0, 2, 6, 10, 24, 3, 34, 9, 8, 12, 14</p>                                     | 10 |

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22-11-2017(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)  
**End Semester Exam**  
Nov-Dec 2017

**Max. Marks: 100**

Class: SYBTech

Name of the Course: Digital Logic Design and Application

Course Code: UITC304

**Duration: 3 hours**

Semester: III

Branch: IT

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. | Question / Short solution / steps / main contents of answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Max. Marks |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q1 (a)       | Solve the following (any 5) (write intermediate steps) <ol style="list-style-type: none"> <li>i. <math>(126.7)_8 = (\quad)_{10}</math></li> <li>ii. <math>(85.408)_{10} = (\quad)_{16}</math></li> <li>iii. <math>(00011001)_{BCD} = (\quad)_{GRAY}</math></li> <li>iv. <math>(100111.1011)_2 = (\quad)_{10}</math></li> <li>v. <math>(94)_{16} = (\quad)_{EXCESS\ 3}</math></li> <li>vi. <math>(131.3)_8 = (\quad)_{16}</math></li> </ol>                                                                                                                                                         | 10         |
| Q1 (b)       | Represent following expressions in canonical SOP as well as canonical POS form. <ol style="list-style-type: none"> <li>i. <math>f(P, Q, R) = \sum m(3, 4, 5)</math></li> <li>ii. <math>Y = AB + A'C' + A'BC</math></li> </ol> <p align="center">OR</p> Solve the following examples <ol style="list-style-type: none"> <li>i. <math>(1101)_2 - (1111)_2</math> (Solve using 2's Complement method)</li> <li>ii. <math>(246)_{BCD} + (38)_{BCD} = (?)_{BCD}</math> (Solve Using BCD Arithmetic)</li> <li>iii. <math>(98.A)_H + (10.6)_H = (?)_H</math> (Solve without changing the base)</li> </ol> | 10         |
| Q2 (a)       | Minimize following expression using Quine McCluskey method and realize the minimized expression using any one type of universal logic gates.<br>$Y(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$                                                                                                                                                                                                                                                                                                                                                                                                 | 10         |
| Q2 (b)       | Minimize the following expression using K Map to get SOP expression and realize using NOR gates only.<br>$F(W, X, Y, Z) = W'X'Y' + X'YZ' + W'XYZ' + WX'Y'$                                                                                                                                                                                                                                                                                                                                                                                                                                         | 10         |
| Q3 (a)       | Design a 4 bit binary to Gray code converter using suitable Demultiplexer.<br><p align="center">OR</p> Design a 1 digit BCD adder using MSI devices.                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10         |

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| Q3 (b) | <p>Design a burglar alarm system for a house having 3 rooms which operates in following way( Show step-wise design procedure)</p> <ul style="list-style-type: none"> <li>• If any two rooms are dark and the laser beam near the door is interrupted by any obstacle, the alarm should ring.</li> <li>• If all rooms are lit, the alarm should not ring.</li> </ul> | 10 |
| Q4 (a) | <p>Design a synchronous counter for generating following sequence using JK flip-flop.<br/>000 – 001 – 011- 010- 110- 111- 101 – 100- 000</p> <p style="text-align: center;">OR</p> <p>Design a 3 bit Johnson counter using D flip-flop. Also draw the labeled timing diagram.</p>                                                                                   | 10 |
| Q4 (b) | <p>Convert following one type of flip-flop to other</p> <ol style="list-style-type: none"> <li>SR to D flip flop</li> <li>JK to T flip flop</li> </ol>                                                                                                                                                                                                              | 10 |
| Q5 (a) | <p>Explain the working of 1 bit magnitude comparator; write a program in VHDL for the same. Write comments for the code.</p>                                                                                                                                                                                                                                        | 10 |
| Q5(b)  | <p>Write a short note on CPLD architecture.</p> <p style="text-align: center;">OR</p> <p>Write a short note on VHDL program format. Explain with example of Full Adder.</p>                                                                                                                                                                                         | 10 |

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24-11-2017(E)

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
Nov – Dec 2017

**Max. Marks: 100**

**Class: SY BTECH**

**Name of the Course: Theory of Computers**

**Course Code: UITC305**

**Duration: 3 Hrs.**

**Semester: III**

**Branch: IT**

**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max. Marks |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| Q 1 (a)      | Design minimized DFA for accepting strings ending with 100 over alphabet (0, 1).                                                                                                                                                                                                                                                                                                                                                                 | 10         |
| Q 1 (b)      | Design Mealy machine to accept all strings ending with aa or bb.                                                                                                                                                                                                                                                                                                                                                                                 | 10         |
| Q2 (a)       | Give formal definition of regular expression. Give R.E. for the following:-<br>(i) Set of all strings over {0, 1} that end with 1 and has no substring 00.<br>(ii) Set of all strings over {0, 1} with even number of 1's followed by odd number of 0's.<br>(iii) Set of all strings over {0, 1} with even length.<br>(iv) Set of all strings over {0, 1} containing at least two 0's.                                                           | 10         |
| Q2 (b)       | State and Prove Pumping Lemma for regular languages and check whether following language is regular or not $L = \{a^n b^n \mid n \geq 1\}$ .<br>OR<br>State and Explain Closure Properties of Regular Languages.                                                                                                                                                                                                                                 | 10         |
| Q3 (a)       | Define CFG and construct a CFG for $a^{2n}b^n$ .                                                                                                                                                                                                                                                                                                                                                                                                 | 10         |
| Q3 (b)       | Convert the following Grammar G to GNF:<br>$G = \{(A_1, A_2, A_3), (a, b), P, A_1\}$ where P consist of following Productions<br>$A_1 \rightarrow A_2 A_3$<br>$A_2 \rightarrow A_3 A_1 \mid b$<br>$A_3 \rightarrow A_1 A_2 \mid a$<br>OR<br>Convert the following grammar to Chomsky Normal Form. Show all the relevant steps briefly.<br>$S \rightarrow bA \mid aB$<br>$A \rightarrow bAA \mid aS \mid a$<br>$B \rightarrow aBB \mid bS \mid b$ | 10         |
| Q4 (a)       | Design PDA for recognizing $L = \{a^m b^n c^{m+n} \mid m, n \geq 1\}$                                                                                                                                                                                                                                                                                                                                                                            | 10         |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| Q4 (b) | <p>Design a PDA corresponding to grammar <math>S \rightarrow aSa \mid bSb \mid \epsilon</math></p> <p>OR</p> <p>For the PDA<br/> <math>(\{q_0, q_1\}, \{0, 1\}, \{0, 1, z_0\}, \delta, q_0, z_0, \phi)</math> where <math>\delta</math> is<br/> <math>\delta(q_0, \epsilon, z_0) = \{(q_1, \epsilon)\}</math><br/> <math>\delta(q_0, 0, z_0) = \{(q_0, 0z_0)\}</math><br/> <math>\delta(q_0, 0, 0) = \{(q_0, 00)\}</math><br/> <math>\delta(q_0, 1, 0) = \{(q_0, 10)\}</math><br/> <math>\delta(q_0, 1, 1) = \{(q_0, 11)\}</math><br/> <math>\delta(q_0, 0, 1) = \{(q_1, \epsilon)\}</math><br/> <math>\delta(q_1, 0, 1) = \{(q_1, \epsilon)\}</math><br/> <math>\delta(q_1, 0, 0) = \{(q_1, \epsilon)\}</math><br/> <math>\delta(q_1, \epsilon, z_0) = \{(q_1, \epsilon)\}</math><br/> Obtain CFG accepted by the above PDA and simplify the CFG and describe the language it accepts.</p> | 10 |
| Q5 (a) | Construct a TM for the language $L = \{w\#w \mid w \in \{0, 1\}^*\}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10 |
| Q5 (b) | <p>Write a short note on (any one)</p> <p>i) Halting Problem</p> <p>ii) Church-Turing thesis</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 10 |